A Directional Connectivity Feature Enhancement Network for Pavement Crack Detection

Zhong Qu, Jian-Dong Wang, and Xue-Hui Yin

***Abstract*— Crack detection is a fundamental effort to ensure road driving safety, aiming to detect potential safety hazards and avoid serious accidents. However, cracks can not be extracted completely and accurately due to problems such as low contrast, high noise, and complex topology of pavement cracks. To address these issues, we propose a directional connectivity feature enhancement network for pavement crack detection. In this network, we build multi-directional enhanced convolution to capture the complex topology of cracks, which is more sensitive to long cracks. To leverage both low-level detail information and high-level semantic information in the network, a novel multi-scale fusion attention is constructed to strengthen the mutual guidance of crack information between channels. The directional connectivity is introducted to establish loss module, which enhances the position and direction information between neighbouring pixels and further refines the crack edge features. To validate the effectiveness and accuracy of the proposed method, we experiment on six publicly available crack datasets, DeepCrack, Crack500, CFD, DCD, EdmCrack600 and DCCE. Compared to other networks, our network achieves 2.2% improvement in ODS and 1.8% improvement in MIoU on the DeepCrack dataset, and 1.2% improvement in ODS and 0.9% improvement in MIoU on the Crack500 dataset. Sufficient experimental results show that our network has better crack detection performance.**

***Index Terms*— Crack detection, multi-directional enhanced convolution, multi-scale fusion attention, directional connectivity.**

I. Introduction

CRACKS are the most common, the most likely to occur, and the early stage of harmful diseases on pavement. Timely detection and maintenance play an influential role in ensuring pavement safety. The degree of damage caused by cracks varies depending on their shape. Therefore, cracks must be detected to capture the morphological features of cracks, which provides a basis for analysis of the cause of cracks, assessment of hazard levels, and selection of repair methods. Early detection methods rely on artificial vision, which requires significant human and time costs, is inefficient, and is prone to missed detections and false detections.

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With the development of computer vision, digital image processing techniques have been used for automatic crack detection, achieving favorable results in specific scenarios. Digital image processing techniques mainly process images through methods such as image thresholding, minimum path, edge detection, wavelet transform, and percolation models [1], [2], [3], [4]. These methods are sensitive to complex pixel backgrounds. Therefore, the crack detection methods for digital image processing techniques have obvious requirements on the image quality, and there are some limitations in the face of complex background noise and different topological structures of cracks, resulting in limited practical application effect.

In recent years, the methods based deep learning have been widely applied to the task of pavement crack detection. Deep learning learns richer and more complex crack features through large-scale data with great accuracy and robustness, and can effectively adapt to crack detection tasks in various scenarios. Effective feature extraction and multi-scale feature information are particularly critical in crack detection with convolutional neural networks. Since the complexity of crack structures and the diversity of multiscale structures, there is still potential for improvement in these two aspects.

Therefore, we propose a novel crack detection method based on directional connectivity and multi-scale feature enhancement. For the long strip cracks, a single square convolution does not fully extract the features and also introduces complex background noise, as shown in Fig. 1. We use a multi-directional strip convolution, paired with an efficient dilated spatial pyramid module, to compensate for the feature extraction capability, enabling the extraction of complex topological crack features and preliminarily ensuring their completeness and accuracy. To leverage the low-level detail information and high-level semantic information of this network, we propose a novel attention mechanism to fuse the relations between channels and enhance the mutual guidance ability of crack detail and semantic information, thereby reducing noise influence and making the features clearer. Aiming at the lack of inter-pixel information and underutilization of crack local detail information in the existing binary classification loss method for single-channel mask labeling, we introduce directional connectivity to establish loss computation module, enhancing the natural positional directional information between pixels. The use of these three methods together enhances the performance of the network, and these methods interact with each other. The first two methods provide as much accurate feature information as possible, while the

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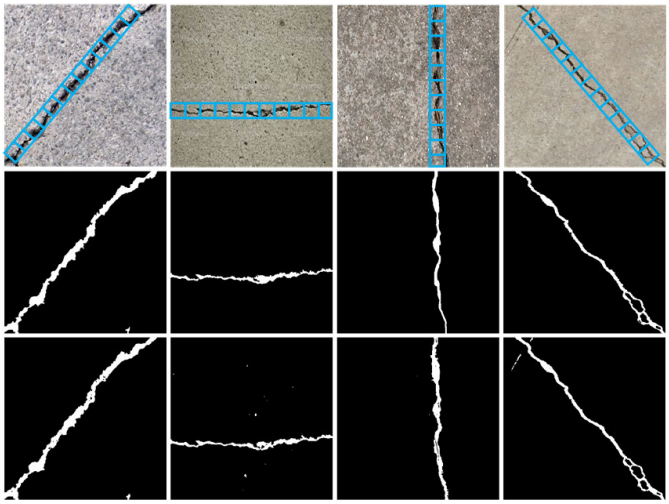

Fig. 1. The shape of the crack in different directions. From top to bottom: Raw Image, Ground truth, Detection result of our proposed method.

second method further refines the fracture information. In this paper, our main contributions are as follows:

1. A multi-directional enhanced convolution module is proposed to capture the crack features of complex topologies. This module focuses on specific long-rang crack features by using strip convolution in different directions, and combines with an efficient dilated spatial pyramid convolution to make up for the extra crack features, making the extracted features more complete and more accuracy.

2. We build a novel multi-scale fusion attention to leverage the local and global spatial location information of crack features to learn interdependence between channels, and adopt appropriate hierarchical crack features for effective fusion, further integrating high-level semantic features with low-level detailed features.

3. The directional connectivity loss is introduced to establish the loss computation, constrain the directional information between neighboring pixels, maintain spatial consistency of crack pixels, and refine crack edges, which integrates features in the same direction from the connectivity information of side outputs and decouples the accurate crack feature information from them, fully utilizing the connectivity between pixels. Meanwhile, we construct a joint loss module using side-output features and side-fused features to jointly supervise the learning of the network.

The subsequent sections in this paper are organized as follows. Sec. II briefly introduces related work. Sec. III describes our proposed network in detail. Sec. IV represents the experimental process and result analysis. Sec. V includes the conclusions of this paper.

II. Related Works

In this section, we briefly introduce related computer vision techniques and methods used to improve the accuracy and effectiveness of detection in crack detection tasks.

A. Feature Extraction

Convolutional neural networks normally use convolution and attention mechanism to capture crack features. In order to obtain global context information in complex crack features, Chen and Lin [5] proposed the hybrid atrous convolutional network (HAC) to distinguish cracks of different sizes by using hybrid connections between different atrous convolution, and Zhou et al. [6] used cascaded multi-scale dilated convolution with four dilated rates to obtain multi-scale information, which grasped the overall feature structure, but also lacked the ability to focus on local details. Therefore, Zhang et al. [7] developed the parallel convolutional-Transformer mixing module to capture the sophisticated local features crucial global context. By effectively adapting the features of different shapes, Dai et al. [8] proposed deformable convolution to break through fixed geometric convolution and perform dynamic spatial position sampling. Cheng et al. [9] introduced an irregular-aware method based on deformable convolution, aiming to design a feature extraction method consistent with crack characteristics. Qi et al. [10] proposed dynamic snake convolution, whose selection is constrained by continuity, making it more effective for elongated features. However, when applied to crack detection, it may be disturbed by a large amount of noise. The position of dynamic sampling is shifted largely, and the topological features of cracks cannot be captured effectively. Huang et al. [11] used information in horizontal and vertical directions and lacking information in other directions. Zhou et al. [12] proposed a novel split and exchange convolution (SE Conv) to exchange information at different scales to fully obtain multi-scale information, and Chen et al. [13] combined multi-scale convolution and hybrid attention to enhance high-level features and utilized spatial and channel relationship features to improve segmentation accuracy. These methods demonstrate the effectiveness of using multi-scale feature extraction in crack detection. Therefore, based on multi-scale information, effectively utilizing dilated convolutions and multi-directional feature information is a promising approach to feature extraction.

B. Feature Fusion

High-resolution shallow features have abundant detailed information, but have large background noise. Low-resolution deep features have abstract semantic information, but are insensitive to details. Therefore, feature fusion at different scales is critical for crack detection. Qu et al. [14] proposed a novel multi-scale convolutional feature fusion method that directly introduced high-level features to low-level features at different stages. In order to further utilize crack information, Yang et al. [15] adopted a progressive multi-scale feature fusion approach to cross-fuse information from high and low levels, and retained rich information. These methods have explored the effectiveness of appropriate fusion methods at different stages for crack detection. To give full play to the guidance ability of different layers, Yang et al. [16] adopted the hierarchical weight method to balance the integration of the high-level semantic information the low-level detailed information from the feature pyramid. In addition to the reasonable selection of layered features, an effective fusion method is also a key step to play a role. Through the use of multi-scale channel attention, Liu et al. [17] proposed shape

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semantic priors module to improve the guidance ability of the high-level to low-level in the global information, while the local information guidance ability was relatively weak and the control of details was lacking. To compensate for the ability to guide local detail features, Zhou et al. [18] used global, horizontal and vertical mixed attention, made use of the global information in space to optimize the details, but didn't exert the global information in the relationship between channels, and Zhang et al. [19] carried out "padding-shifting-cutting" operation on the feature graph to obtain the distant dependency. Therefore, we will adopt multi-level feature fusion in line with the backbone network, make full use of the guidance ability of global information in space and channel, and further exert the feature advantages extracted from the backbone network.

C. Connectivity Modeling

The connectivity of feature map pixel is a multi-dimensional binary vector that indicates whether it is connected to its neighboring pixels. Doersch et al. [20] introduced the concept of similarity in spatial context, and divided random sampling area to supervise the learning of the similarity between pixels. How to use the connectivity between pixels is critical. By introducing directional connectivity into channel attention, Li et al. [21] obtained the weights of pixels in the same direction, and then guided the markings in urban road images to maintain consistency. In addition to using directional connectivity information in the network, Yang et al. [22] introduced connectivity modeling into the loss function, using the connectivity relationship between pixels to jointly constrain the convergence of the network. Frequently, a single connectivity loss is not sufficient to support the feature learning ability of the network and additional losses may need to be used as guiding functions. Rasti et al. [23] combined the loss of segmentation and connectivity to optimize the loss function, so as to obtain better segmentation results. Meanwhile, connectivity can also be applied in feature information extraction. Yang and Farsiu [24] decoupled information in specific subdirection from the direction space of connectivity features at the higher level, acting on connectivity features at the lower level, so that connectivity features are consistent in the same direction. Since this structure is derived from the detection of medical retinal fluid, in a small amount of data, it is necessary to attach the entire network to the connectivity to maintain the consistency. However, in crack detection, due to the existence of the complex background noise, the connectivity affects the consistency between pixels. We need to build the network that matches the crack characteristics in the backbone, decouples the correct crack information from the side-output connectivity feature maps, and takes full advantage of connectivity relationships with joint loss to refine the crack features.

III. Proposed Methods

In this section, we introduce the proposed network in detail, including multi-directional enhanced convolution module, multi-scale fusion attention, and directional connectivity loss module. By using the above three methods, we can effectively improve the accuracy and robustness of crack detection.

A. Network Structure

The crack detection task is a two-class semantic segmentation problem. On the basis of preserving the classification recognition capability, the positional relationship between pixels is sufficiently utilized. Our proposed network consists of three parts: multi-directional enhanced convolution module, multi-scale fusion attention, and directional connectivity loss module, as shown in Fig. 2. Inspired by DeepCrack [25], we adopt the same backbone network architecture with it, and use the multi-directional enhanced convolution module as the basic module of the backbone network to capture the complex topology of cracks and provide as complete crack information as possible for subsequent methods. The overall backbone network is divided into five stages. In the first three stages, each stage includes two basic modules, and in the last two stages, each stage includes three basic modules. We consider that the high-resolution crack feature of the lower layer contains more detailed information, but there is more redundant noise. Therefore, when too many basic modules are used to learn in the first three stages, complex noises may be embedded into the feature information, which is not conducive to the subsequent feature extraction and the fusion of side outputs. The last two low-resolution stages need to extract rich semantic information, so it is necessary to further strengthen the learning ability of features. Five stages produce side outputs $(S_1, S_2, S_3, S_4, S_5)$. To better explain the characteristics of the crack features at each stage and the use of subsequent fusion methods, we divide the side outputs into three parts, (S_1, S_2), S_3, (S_4, S_5). In the first part, a lot of crack detail information is retained, and there is inevitably a large amount of background noise. The second part is a moderate block, where both detailed and semantic information are represented to some extent. The third part mainly presents deep semantic information. Therefore, to leverage both high-level semantic information and low-level detailed information, S_2 and S_3 are fused, and S_3, S_4 and S_5 are fused. S_2 and S_3 are two adjacent stages, which have a certain connection in feature information, and S_3 contains limited semantic information, so it is more appropriate guide S_2. If high-level semantic information is directly used, the information extraction may be biased at the lower level. The integration of S_3, S_4 and S_5 strengthens the guidance of high-level semantic information, so as to generate a buffered fusion mode, which can better maintain the integrity of features. Then the spatial position information of the crack is obtained through the multi-scale fusion attention to select effective crack information. The two fused side outputs become the new S_4, S_5. The side outputs of the original S_1, S_2 and S_3 of the backbone network and new S_4, S_5 form the last five side outputs. The side outputs are finally constrained by the directional connectivity loss module to enhance the capability ability of the network and refine the edge details.

B. Multi-Directional Enhanced Convolution Module

Crack feature extraction is the most basic step in crack detection tasks. The quality of feature extraction largely

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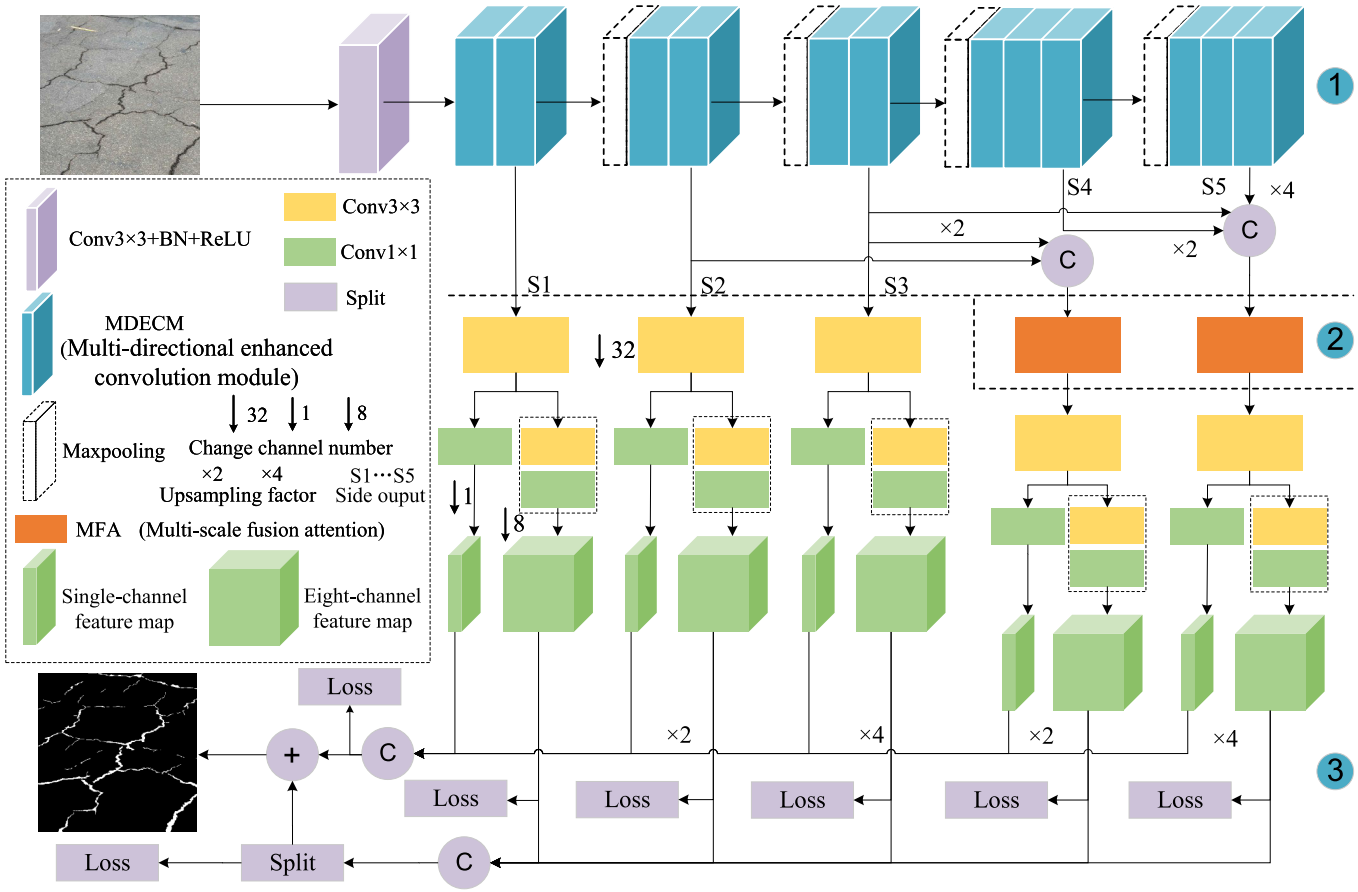

Fig. 2. The architecture of our proposed method. Part one: MDCM, extract the complex topological structure features of cracks. Part two: MFA, fuse multi-scale features and learn dependencies between channels of feature map. Part three: Connectivity Loss module, establishe a relationship between crack pixel position and direction to constrain model learning. Among them, Loss does not explicitly distinguish between connectivity loss and single-channel binary cross-entropy loss.

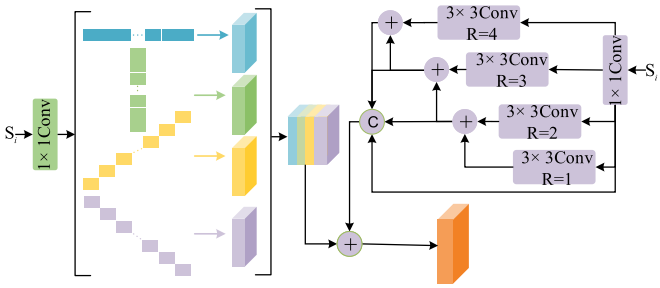

Fig. 3. Multi-directional Enhanced Convolution Module(MDECM). We combine strip convolutions with four different directions (horizontal, vertical, left diagonal, right diagonal) and an efficient dilated spatial pyramid with rate = 1,2,3,4 to extract crack features. S_i is the input of the module, which equals the side output value of the previous stage.

determines the performance of subsequent methods. The complex topological structure of cracks has a large influence on feature extraction, and it is not common to extract complete and accurate crack features. Inspired by Strip Pooling [26], Coat [27], and ESPNet [28], the multi-directional enhanced convolution module (MDECM) is proposed as the basic module of the backbone network. The multi-directional enhanced convolution module combines directional mixed strip convolution (DMSC) and efficient dilated spatial pyramid convolution (ESP) to jointly capture the complex topological structure features of cracks, as shown in Fig. 3.

The directional mixed strip convolution (DMSC) is sensitive to long strip cracks and effectively extract the corresponding features. This module uses 1D strip convolutions in four directions to extract crack features, then which are fused through channel concatenation operations to obtain directional features. In order to have the same receptive field size as the 3×3 convolution, the 1D strip convolution uses a convolution kernel size of 9. Strip convolutions of 1×9 and 9×1 are used in the horizontal and vertical directions respectively. The left diagonal and right diagonal strip convolutions essentially use the same convolutional kernel as the horizontal and vertical convolutional kernels. By transforming the feature map, the pixels at the sampled positions in the diagonal direction are transformed into horizontal or vertical directions, and the features in the diagonal direction are extracted according to the computational behavior of the 1D strip convolution, as shown in Formula (1). The efficient dilated spatial pyramid convolution is to enhance the global crack features and avoid using only strip convolution to ignore the additional cracks information. Due to the non-uniqueness of the crack feature shapes, most features present a long-range topological structure, but there are wider or other irregular crack features. This module uses four parallel dilated convolutions with dilate

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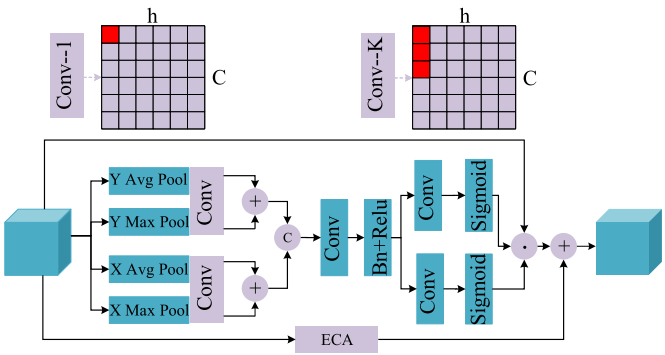

Fig. 4. X is horizontal and Y is vertical. Conv operations include Conv–1 and Conv–k in the pink block diagram. Conv–1 is the convolution operation using 1 × 1 in MFA-A. Conv–k is the convolution operation using K×1 in MFA-B. C is the number of channels and h is the height of the feature map, taking the X-direction as an example.

rates of [1, 2, 3, 4] to capture feature information of different scales. To avoid the grid hole problem caused by directly fusing parallel dilated convolutions, this module adds the results of layer-wise dilated convolutions to compensate for gaps in receptive fields. The module is implemented using depthwise separable group convolutions to improve running speed and reduce the number of parameters. The final output is element-wise addition of the results from these two modules. The formulas for this method are as follows,

$$Z_D[i, j] = \sum_{l=-k}^{k} x[i + D_h \cdot l, j + D_w \cdot l] \cdot w[k - l], \tag{1}$$

$$F = CBR(Cat(Z_{D(0,1)}, Z_{D(1,0)}, Z_{D(1,1)}, Z_{D(-1,1)})) + ESP, \tag{2}$$

where $X \in \mathbb{R}^{H\times W\times C}$ is the input feature map, and $Z_D \in \mathbb{R}^{H\times W\times C'}$ represents the result of strip convolution. $w \in \mathbb{R}^{2k+1}$ is a strip convolution filter of size $2k + 1$, and setting $k = 4$ gives the strip convolution have a receptive field of 9. $D = (D_h, D_w)$ is the direction of filter w and is the direction vector of strip convolution, including horizontal (1, 0), vertical (0, 1), left diagonal (−1, 1) and right diagonal (1, 1). CBR includes the convolution, batchnormalization and relu operation, and Cat represents the concatenate operation. ESP is the output of the efficient dilated spatial pyramid convolution.

C. Multi-Scale Fusion Attention

To fully utilize the potential information between feature map channels and enhance hierarchical crack feature information, inspired by ECA [29], CBAM [30], and CoordAttention [31], we propose multi-scale fusion attention (MFA), as shown in Fig. 4. MFA includes two structures: MFA-A and MFA-B. ECA and CBAM explore the relationship between feature channels, but focus more on global range information and cannot obtain local long range dependencies, thereby ignoring the importance of position information. CoordAttention embeds position information into channel attention and integrates spatial information. MFA further enhances the utilization of local and global spatial position information.

MFA-A uses two parallel branches to extract spatial position information, including vertical and horizontal pooling branches and ECA branches. Among them, the pooling branch also uses a parallel method to focus on the long distance directional spatial information, while ECA focuses on the information of the entire spatial range. The pooling branch uses 1D average pooling and 1D max pooling to extract directional spatial position information. Let $X(c, h, w)$ be used as the input feature map, where c, h, and w represent channel number, length and width respectively. The output results through horizontal and vertical pooling operations are $X_h(c, h, 1)$, $X_v(c, 1, w)$ respectively (regardless of the specific type of pooling). The horizontal and vertical pooling results are each followed by a 1×1 convolution to obtain information between channels. Then the vertical feature map is recoded as $X_v(c, w, 1)$, and both X_h and X_v are concatenated in the second dimension to become $X(c, h+w, 1)$. This output result integrates information between positions and channels through 1×1 convolution, batchnormalization and relu operations. After that, X is separated into X_h and X_v according to the values of h and w. The input feature map is respectively multiplied by the position weights through applying the sigmoid operation to X_h and X_v as the pooling branch result. The outputs of the ECA branch and the pooling branch perform element-wise addition as the final result.

MFA-B is a modification based on MFA-A. The difference is the convolution operation after pooling. MFA-A uses a 1×1 convolution to explore the dependences between channels at a certain pixel position, but the relationships between channels obtained in this way are relatively simple and do not properly reflect the mutual relationships between channels. Inspired by the ECA of constructing inter-channel dependencies, the 1D K×1 convolution is used to learn the relationship between channels. K [29] is calculated by Formula (3), (4). The pooling results X_h and X_v are re-encoded as $X_h(1, c, h)$, $X_v(1, c, w)$ to perform the K×1 convolution operation. The formulas for this method are as follows,

$$t = \lfloor |(\log(c, 2) + 1)/2| \rfloor, \tag{3}$$

$$K = t + t \bmod 2, \tag{4}$$

$$Pooling_d = Conv_k(AvgPool_d(X)) + Conv_k(MaxPool_d(X)), \tag{5}$$

$$X_h, X_v = Split(CBR(Cat(Pooling_h, Pooing_v))), \tag{6}$$

$$Output = X * Sigmoid(Conv(X_h)) * Sigmoid(Conv(X_v)) + ECA, \tag{7}$$

where c is the number of input channels. X is the input feature map, $Pooling_d$ is the result of pooling in d direction (horizontal and vertical). $Conv_k$ is the K×1 convolution operation. $Split$ is the separation operation, outputting X_h and X_v. $Sigmoid$ is the activation function that produces position weights. ECA is the result of the efficient channel attention branch.

D. Directional Connectivity Loss Module

There are eight neighboring pixels around a single pixel, and these pixels have inherent direction position attributes.

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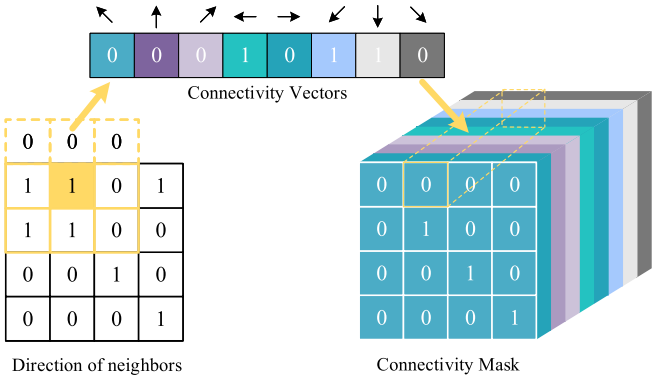

Fig. 5. An eight-channel connectivity feature map generated by eight neighboring pixels around a pixel.

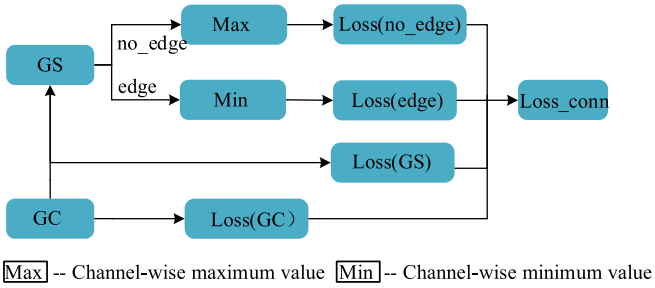

Fig. 6. Connectivity loss. GC is an eight-channel connectivity feature map. GS is the connectivity feature map produced by GC through the bilateral voting mechanism. Loss is the binary cross-entropy loss.

An eight-channel connectivity map contains class and directional information about cracks, with each channel representing a specific direction. When two neighboring pixels are both crack pixels, it means that they are connected (0 represents nonconnected, 1 represents connected), as shown in Fig. 5.

The single-channel label mask focuses on pixel categories and does not exploit the naturally existing directional relationships between pixels. Therefore, we introduce directional connectivity loss [22] and build the directional connectivity loss module, as shown in Fig. 6.

Directional connectivity loss [22] includes the eight-channel connectivity loss, the bilateral voting loss, the no_edge loss, and the edge loss. These losses essentially use binary cross-entropy loss. The eight-channel loss is calculated based on channels and pixels individually. The bilateral voting loss calculate connectivity map after passing the bilateral voting mechanism. The bilateral voting mechanism aims to maintain consistent connectivity between two neighboring pixels as much as possible. The positions of two neighboring pixels are interrelated: if pixel A and pixel B are neighbors, pixel A assigns pixel B to position j in its connectivity vector, and correspondingly, pixel B assigns pixel A to position $9 - j$ in its connectivity vector. The connectivity between pixel A and pixel B should be the same. Therefore, the original value is replaced by the result of multiplying the two values, respectively. The no_edge loss is used in non-edge pixel areas. The eight-channel features after bilateral voting are maximized along the channel dimension and converted to single channel for loss calculation. The edge loss is applied to the edge pixel areas to suppress the error probability in the eight channels, which makes the edge details as correct as possible. The eight-channel label mask in the edge pixel areas is mixed vectors of 0 and 1, while the non-edge pixels are single vectors of all 0 or all 1, as shown in Fig. 7.

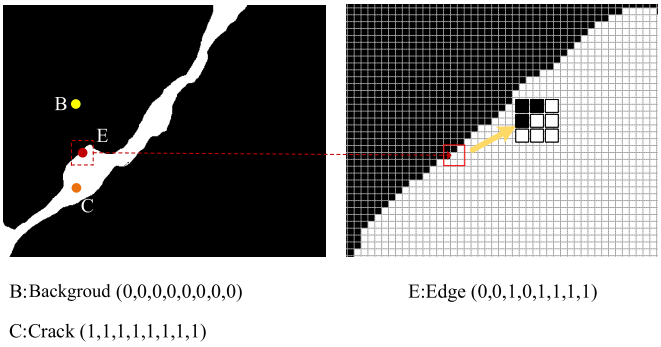

Fig. 7. Connectivity vectors of edge crack region and non-edge crack region.

The directional connectivity loss module combines the directional connectivity loss and the standard single-channel binary cross-entropy loss to constrain the model and improve the accuracy and convergence speed of crack detection. The directional connectivity loss not only constrains the crack class identification to a certain extent, but also better fits the directional topology of cracks. The standard single-channel binary cross-entropy loss is maintained to adjust the learned features on the backbone network so that the features do not deviate too much due to the eight-channel loss, thus accelerating the directional connectivity loss to learn better features.

The directional connectivity loss module uses the directional connectivity branch and the single-channel binary classification branch in parallel. First, the channel numbers of the five side outputs (S_1, S_2, S_3, S_4, S_5) are reduced to 32 by a 3×3 convolution. They do not directly reduce to eight-channel or single-channel to avoid the mutual interference between the two losses. The intermediate state method is able to learn the feature maps required by both losses in a comprehensive and abstract way. The directional connectivity branch reduces the channel numbers of the feature map from 32 to 8 by the 3×3 convolution, and then re-encodes the eight-channel feature map using the 1×1 convolution. The output feature map is upsampled to the input feature map size through bilinear interpolation. It is worth noting that we wish the side outputs to only learn more eight-channel connectivity relations without emphasizing the learning of binary categories, hence the no_edge loss is not used in the side for directional connectivity. To enhance the directional information of the eight channels, the eight channels of the five side outputs are divided from each other, and the connectivity channels in the same direction are combined, and 1×1 convolution is used in each direction to adjust the channel number from 5 to 1. By concatenating the eight single-channels, the connectivity relations in each direction can be more effectively obtained, facilitating fusion loss calculation, as shown in Fig. 8. The single-channel binary classification branch directly reduces the

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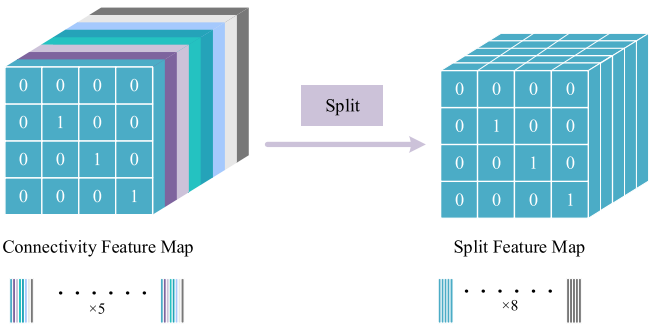

Fig. 8. Fuse the features in the same direction from the five side eight-channel connectivity output feature maps.

number of channels from 32 to 1 and then upsamples the side outputs using bilinear interpolation, which are concatenated in the channel dimension and encoded into the single channel for fusion loss. The output results of the directional connectivity branch and the single-channel binary classification branch are added as the final output of the model. The formulas for this method are as follows,

$$S_i' = Recode_e(Conv(S_i)), \tag{8}$$

$$S_i'' = Recode_o(Conv(S_i)), \tag{9}$$

$$C_a = Conv(Cat(S_{1,a}' + S_{2,a}' + S_{3,a}' + S_{4,a}' + S_{5,a}')), \tag{10}$$

$$F_b = Cat(C_1, C_2, \ldots, C_8), \tag{11}$$

$$F_c = Conv(Cat(S_1'', S_2'', S_3'', S_4'', S_5'')), \tag{12}$$

where S_i' is the side connectivity map, S_i'' is the side single-channel binary classification map. $Recode$(e-connectivity branch, o-single-channel branch) is the operation of recoding the side output and adjusting the channel dimension. C_a is the fusion of features in the same direction of the pixel, and a is a certain direction. F_b is the result of the fusion of side connectivity maps, and F_c is the result of the fusion of side single-channel classification maps. Cat is the concatnate operation and $Conv$ is the convolution operation.

E. Loss Function

The crack detection task is a binary classification problem that includes both crack pixels and no-crack pixels. For deep supervised training, the binary cross-entropy loss shows great performance. Let $G(x, y) \in (0, 1)$ is the ground truth label of pixel (x, y) corresponding to the training set, and $S(x, y) \in (0 \sim 1.0)$ is the probability of network prediction. The loss function we use is as follows,

$$\begin{aligned} L_{bce}(S, G) = -\sum_{(x,y)} [&w_0 * G(x, y) * \log S(x, y) \\ &+ w_1 * (1 - G(x, y)) * \log(1 - S(x, y))], \end{aligned} \tag{13}$$

where we use a weighted factor w [25] to balance the loss calculation. Due to the uneven distribution of crack pixels and non-crack pixels, the simple cross-entropy loss may cause training difficulties with the saturation behavior of the activation function. So we set $w_0 = \frac{G-}{G+}, w_1 = 1.0$. $G+$ is all crack pixels in the training image and $G-$ is all non-crack pixels in the training image. We introduce the directional connectivity loss to learn the relationship between neighboring pixels, aligning with crack morphological features, refining the edge features, and computing by transforming the single channel label mask to the eight-channel label mask. These loss functions are as follows,

$$\begin{aligned} &L_{conn}(C, \tilde{C}, Gs, Gc) \\ &\quad = \alpha * L_{con_map}(C, Gc) \\ &\qquad + \beta * L_{bv_map}(\tilde{C}, Gc) \\ &\qquad + L_{edge}(\tilde{C}, Gs) + L_{no_edge}(\tilde{C}, Gs) \\ &\quad = \alpha * L_{bce}(C, Gc) + \beta * L_{bce}(\tilde{C}, Gc) \\ &\qquad + L_{edge}(\tilde{C}, Gs) + L_{no_edge}(\tilde{C}, Gs), \end{aligned} \tag{14}$$

$$\begin{aligned} L_{edge}(\tilde{C}, Gs) &= L_{bce}(1 - min\{\tilde{C}_i(x, y)\}_{i=1}^{8}, Gs(x, y)) \\ &= L_{bce}(1 - min\{\tilde{C}_i(x, y)\}_{i=1}^{8}, \mathbf{1}) \\ &= L_{bce}(min\{\tilde{C}_i(x, y)\}_{i=1}^{8}, \mathbf{0}), (x, y) \in P_{edge}, \end{aligned} \tag{15}$$

$$\begin{aligned} L_{no_edge}(\tilde{C}, Gs) &= L_{bce}(max\{\tilde{C}_i(x, y)\}_{i=1}^{8}, Gs(x, y)), \\ &\quad (x, y) \notin P_{edge}, \end{aligned} \tag{16}$$

where Gs is single-channel ground truth label. Gc is the connectivity map ground truth label (eight-channel label) from the single-channel label. C is the prediction result of the connectivity map (eight-channel probability map). $\tilde{C}$ is the connectivity prediction map after bilateral voting mechanism. α and β are weight factor. L_{conn} is the connectivity loss. L_{con_map} is the eight-channel connectivity loss. L_{bv_map} is the bilateral voting loss. L_{edge} is the edge loss. L_{no_edge} is the no_edge loss.

$$L_{conn_side} = \sum_{i=1}^{n=5} \alpha^i * L_{conn_no}(S_i', \tilde{S}_i', Gs, Gc), \tag{17}$$

$$L_{conn_fuse} = L_{conn}(F_b, \tilde{F}_b, Gs, Gc), \tag{18}$$

$$L_{c_fuse} = L_{bce}(F_c, Gs), \tag{19}$$

$$L_{total} = L_{conn_side} + L_{conn_fuse} + L_{c_fuse}, \tag{20}$$

our loss is the sum of the side connectivity loss(L_{conn_side}) with weight factor α^i, the fused connectivity loss (L_{conn_fuse}), and the fused single-channel loss (L_{c_fuse}). L_{conn_no} is L_{conn} without L_{no_edge}. The other symbols have the same meaning as in the previous section.

IV. Experimental Results And Analysis

In this section, we detail the implementation and training process of the proposed network, including experimental settings, datasets, evaluation metrics, comparison methods, analysis of experimental results, and ablation experiments. Fig. 9 and Fig. 10 show the crack detection results of the fourteen methods on different datasets. It is worth noting that our method can more effectively identify the complex structure of cracks and avoid the interference of background noise.

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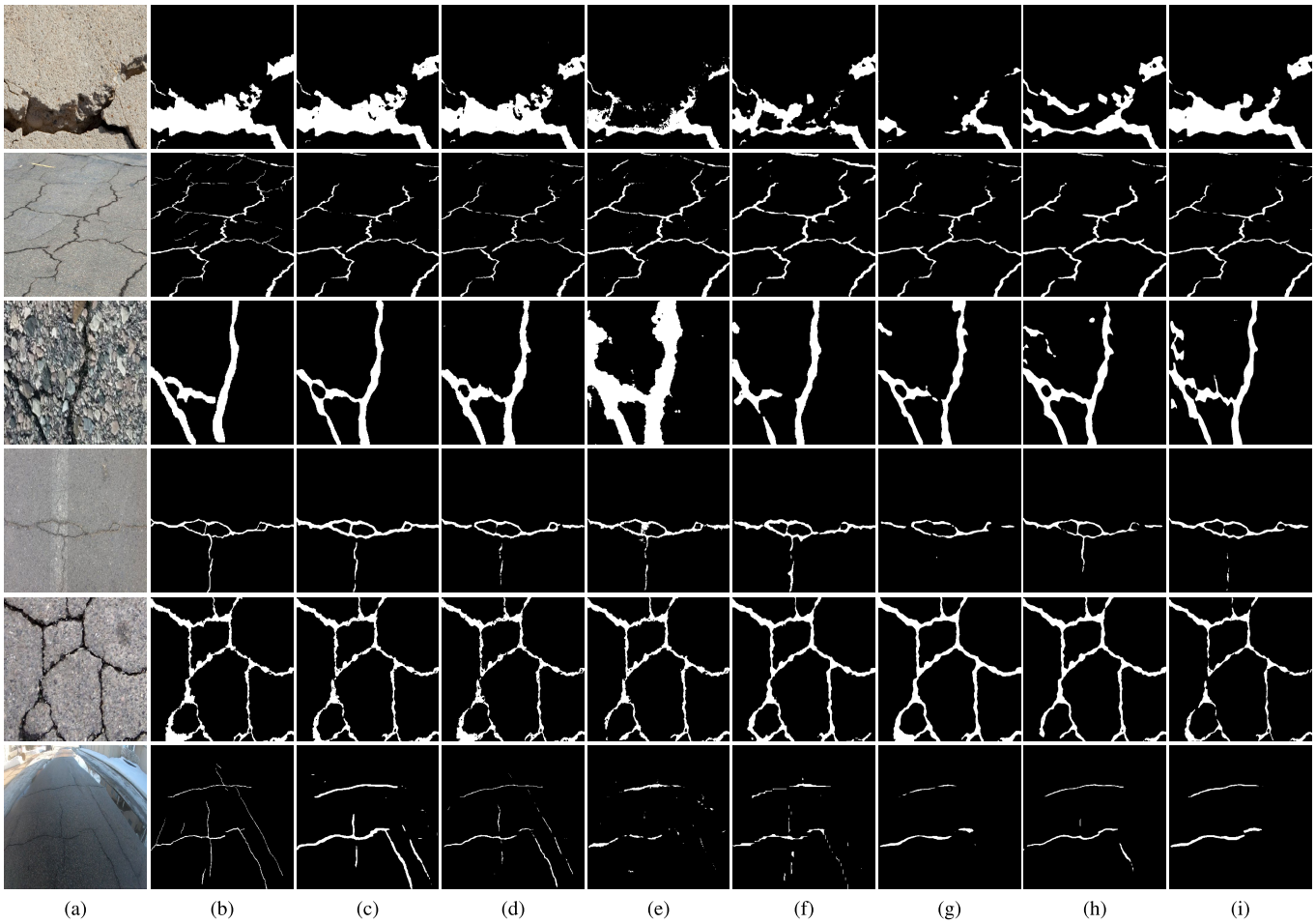

Fig. 9. The visualization of detection results of different methods selected from five datasets. (a) Raw Image; (b) Ground truth; (c) Ours; (d) U-Net; (e) SegNet; (f) DeepLab3+; (g) Crackformer; (h) Dconn-Net; (i) DeepCrack. From top to bottom are DeepCrack, DeepCrack, Crack500, CFD, DCD, EdmCrack600.

A. Implementation Details

The proposed network model uses the Pytorch deep learning framework. The side output loss weight of the model is [0.5, 0.75, 1.0, 0.75, 0.5]. During training, the model uses a stochastic gradient descent (SGD) optimizer with the momentum (0.9) and weight decay (2e - 4) to update network parameters. The learning rate (1e - 4), iterative strategy (Step), 10 times decrease in learning rate per 40 epochs. The model is trained for 200 epochs with the batch size of 1, and experiments are conducted on an NVIDIA A100-PCIE-40GB server.

B. Datasets

1) DeepCrack [25]: This dataset includes 537 pavement crack images, selecting complex cracks with uneven intensity in different scenes. The image size is 544×384 and we adjust the image size to 256×256 for training. We divide the dataset into three parts, including a training set of 300 images, a test set of 187 images, and a verification set of 50 images.

2) Crack500 [16]: A public annotated complex crack dataset, including 3368 crack images, which is cropped from 500 crack images taken by mobile phones. This dataset is mostly long cracks. Among them, 1896 images are used for training and 1472 images are used for testing. The image scale is 640×360, which we adjust to a size of 256×256 for training.

3) CFD [4]: It contains 118 annotated images of concrete pavement cracks, with a size of 480×320. These images have uneven illumination and noise such as shadows and lane lines. Most of the cracks in the image are tiny that are difficult to detect. This dataset is used to evaluate our model.

4) DCD: To additionally confirm the performance of our method, we combine of our datasets together, including DeepCrack and CFD. We call it DCD, which contains 305 pavement cracks. All images are used to evaluate our model.

5) EdmCrack600 [32]: The dataset consists of 600 images of pavement cracks with different lighting and shadows taken from roads in Edmonton, Canada. The images scale is 1920×1080. Among them, 400 images are used for training and 200 images are used for testing.

6) DCCE: To simulate various pavement conditions in real-world engineering environments, we select 140 images from the EdmCrack600, DeepCrack, Crack500, and CFD datasets for training and 70 images for testing. These images exhibit characteristics such as transverse, longitudinal, mesh-like, and irregular crack shapes; complex pavement conditions including high exposure, shadows, and occlusions; different capture angles, such as overhead and frontal views; and various pavement materials, including concrete and asphalt.

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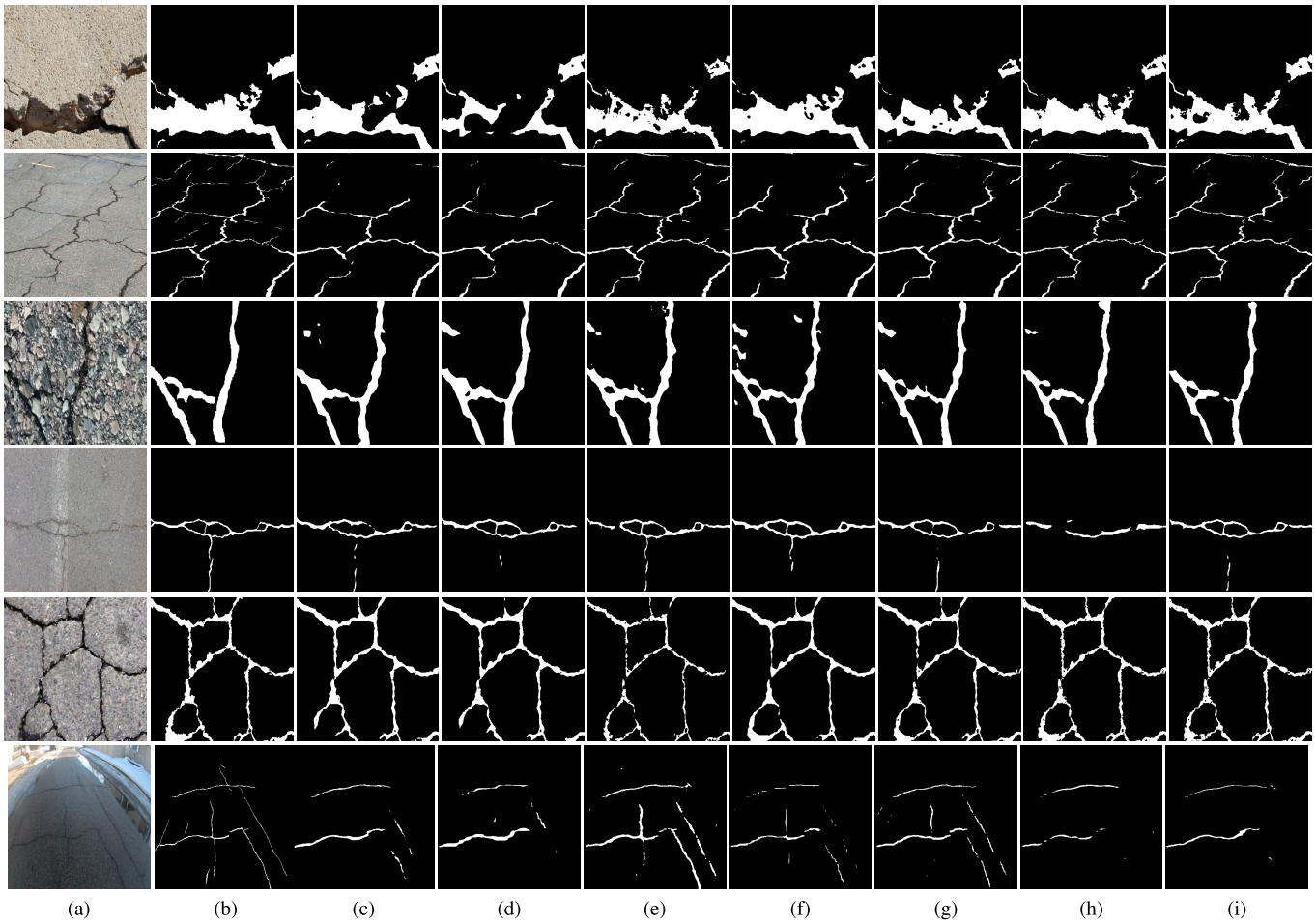

Fig. 10. The visualization of detection results of different methods selected from five datasets. (a) Raw Image; (b) Ground truth; (c) DeepCrackT; (d) FPHBN; (e) CrackSegNet; (f) MSFC; (g) PAF-Net; (h) DSC-Net; (i) CMUNeXt. From top to bottom are DeepCrack, DeepCrack, Crack500, CFD, DCD, EdmCrack600.

C. Comparsion Methods

1) U-Net [33]: U-Net nakes a breakthrough in medical image segmentation by adding a skip layer to the encoder and decoder architecture.

2) SegNet [34]: SegNet is a classical encoder-decoder network structure. The maximum pooling of encoder at each stage guides non-linear upsampling of the decoder.

3) DeepLab3+ [35]: deepLab3+ uses a simple and efficient decoder to refine segmentation and introduces deep separable convolution into atrous space pyramid pooling.

4) Crackformer [36]: Crackformer is a Transformer network for fine-grained crack detection that uses a novel attention to extract contextual information across feature channels.

5) Dconn-Net [24]: Dconn-Net uses directional connectivity modeling, including decoupling, tracking, and utilizing the directional information of the entire network, showing that unraveling the directional subspace from the shared latent space can enhance feature expression.

6) DeepCrack [25]: The backbone of DeepCrack consists of the first 13 convolution layers in the VGG16 network. All side outputs are concatenated to produce the final fusion result.

7) DeepCrackT [37]: The DeepCrackT network is built on the encoder-decoder architecture. The features generated in the encoder and decoder networks are merged pairwise at the same scale.

8) FPHBN [16]: FPHBN integrates the high-level contextual semantic information into the low-level detailed information through feature pyramid and balances the contributions of the two through reweighting.

9) CrackSegNet [38]: CrackSegNet is an improved encoder-decoder architecture based on SegNet. The encoder learns crack features hierarchily, and the decoder upsamples and maps features.

10) MSCF [39]: MSCF proposes an asymmetric encoding and decoding structure to capture multi-scale contextual information through efficient attention module branches at different stages of the network.

11) PAF-Net [15]: PAF-Net is a progressive multi-scale feature fusion method that gradually fuses high-level and low-level information to preserve more rich information.

12) DSC-Net [10]: DSC-Net proposes a dynamic snake convolution to accurately capture the features by adaptively focusing on slender and tortuous local structures.

13) CMUNeXt [40]: CMUNeXt is a light segmentation network that leverages large kernel and inverted bottleneck design to thoroughly mix distant spatial and location information, efficiently extracting global context information.

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D. Evaluation Metrics

To evaluate the performance of these methods, we utilize different evaluation metrics for analysis. Since crack detection is a pixel-level classification task, we introduce Precision(P), Recall(R), F1-score($F1$). These formulas are as follows,

$$P = \frac{TP}{TP + FP}, \tag{21}$$

$$R = \frac{TP}{TP + FN}, \tag{22}$$

$$F1 = \frac{2 \times P \times R}{P + R}, \tag{23}$$

where TP means true positive, FP means false positive and FN means false negative. Due to the limitation of Precision and Recall, we focus on the $F1$ criteria. ODS (Optimal Dataset Scale) is the best $F1$ on the dataset with a fixed threshold, OIS (Optimal Image Scale) is the aggregate $F1$ on the dataset with the best threshold of each image.

$$ODS = max(\frac{2 \times P_t \times R_t}{P_t + R_t}), \tag{24}$$

$$OIS = \frac{1}{N_{img}} \sum_{i}^{N_{img}} max(\frac{2 \times P_t^i \times R_t^i}{P_t^i + R_t^i}), \tag{25}$$

where t = (0.00,0.01,...,0.99) means the selected threshold value, i represents the image currently calculated, N_{img} is the number of dataset image, P_t and R_t mean that the precision and recall value at threshold t. Meanwhile, crack detection is also a binary segmentation task. Therefore, we introduce three semantic segmentation metrics, which are pixel accuracy (PA), mean pixel accuracy (MPA), and mean intersection over union ($MIoU$).These formulas are as follows,

$$PA = \frac{\sum_{i=0}^{n} P_{ii}}{\sum_{i=0}^{n} \sum_{j=0}^{n} P_{ij}}, \tag{26}$$

$$MPA = \frac{1}{n+1} \sum_{i=0}^{n} \frac{P_{ii}}{\sum_{j=0}^{n} P_{ij}}, \tag{27}$$

$$MIoU = \frac{1}{n+1} \sum_{i=0}^{n} \frac{Pii}{\sum_{j=0}^{n} P_{ij} + \sum_{j=0}^{n} P_{ji} - P_{ii}}, \tag{28}$$

where n(n = 1) denotes the number of class, P_{ii} means that the pixel classified as i is predicted to be i, and P_{ij} means that the pixel with classified as i is predicted to be j.

The Floating-Point Operations ($FLOPs$) and Parameter numbers ($Params$) are the key metrics to measure the complexity of the model. The $FLOPs$ and $Params$ of each convolution layer can be calculated as follows,

$$FLOPs = 2HW(C_{in}K^2 + 1)C_{out}, \tag{29}$$

$$Params = (C_{in}K^2 + 1)C_{out}, \tag{30}$$

where H and W represent the height and weight of the feature map, K is the size of convolution kernel. C_{in}, C_{out} indicate the channel number for input and output feature map. In addition, we use Frames Per Second (FPS) to evaluate the running speed.

TABLE I

EVALUATION METRICS OF COMPARED METHODS ON DEEPCRACK

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
U-Net [33]	0.843	0.851	0.847	0.879	0.987	0.922	0.860
SegNet [34]	0.812	0.824	0.818	0.868	0.984	0.908	0.838
DeepLab3+ [35]	0.789	0.813	0.801	0.848	0.982	0.901	0.825
Crackformer [36]	0.830	0.788	0.808	0.848	0.984	0.890	0.831
Dconn-Net [24]	0.826	0.799	0.812	0.842	0.984	0.891	0.833
DeepCrack [25]	0.855	0.850	0.853	0.878	0.987	0.922	0.865
DeepCrackT [37]	0.844	0.819	0.831	0.860	0.985	0.906	0.848
FPHBN [16]	0.839	0.806	0.822	0.842	0.985	0.900	0.841
CrackSegNet [38]	0.813	0.855	0.834	0.886	0.985	0.923	0.850
MSCF [39]	0.830	0.813	0.821	0.856	0.984	0.903	0.840
PAF-Net [15]	0.844	0.865	0.855	0.883	0.987	0.929	0.866
DSC-Net [10]	0.849	0.833	0.841	0.870	0.986	0.913	0.856
CMUNeXt [40]	0.853	0.854	0.854	0.872	0.987	0.924	0.866
Ours	**0.863**	**0.891**	**0.877**	**0.899**	**0.989**	**0.942**	**0.884**

E. Experimental Results

1) Results on DeepCrack [25]: As shown in Fig. 11(a), our method presents the highest peak in the P-R curve compared to the other methods, which means that our network obtain the best precision and recall, indicating the superiority of our network. The quantitative experiments in Table I show that the optimal ODS result of 0.877 is obtained when the threshold is set to 0.24. Among them, Crackformer has lower ODS as it is only suitable for the detection of elongated cracks, while the cracks in DeepCrack dataset are more complex, and the next best method to our method is PAF-Net with progressive fusion method. Our method performs better than Dconn-Net, which also uses directional connectivity. In Fig. 9, we find that Dconn-Net can extract cracks in a basically complete way. However, due to the influence of background noise, the cracks cannot be well located, and extra useless information is often extracted and non-edge crack information is lost. Since Dconn-Net fully relies on connectivity, it decouples the feature information from the connectivity map. The advantage of this approach is that the connectivity can be fully utilized, but complex noise can affect the connectivity between pixels, resulting in poor performance. Our approach uses connectivity only in the lost modules after side fusion. This backbone network and fusion method are more consistent with the characteristics of the crack, and thus avoid possible connectivity bias.

2) Results on Crack500 [16]: As shown in Fig. 11(b), our method performs better than other methods on the Crack500 dataset, indicating that our method has advantages for long cracks. The quantitative experiments in Table II show that the optimal ODS result of 0.753 is obtained when the threshold is set to 0.39. Crackformer is suitable for the Crack500 dataset and shows some improvement, where the performance of FPHBN, CrackSegNet and MSCF are relatively close. DSC-Net adaptively samples pixels and the CMUNeXt effectively mixes long-range spatial and positional information, and they are able to adequately extract long cracks in the Crack500 dataset.

3) Results on CFD [4]: As shown in Fig. 11(c), our method outperforms other methods. The quantitative experiments in

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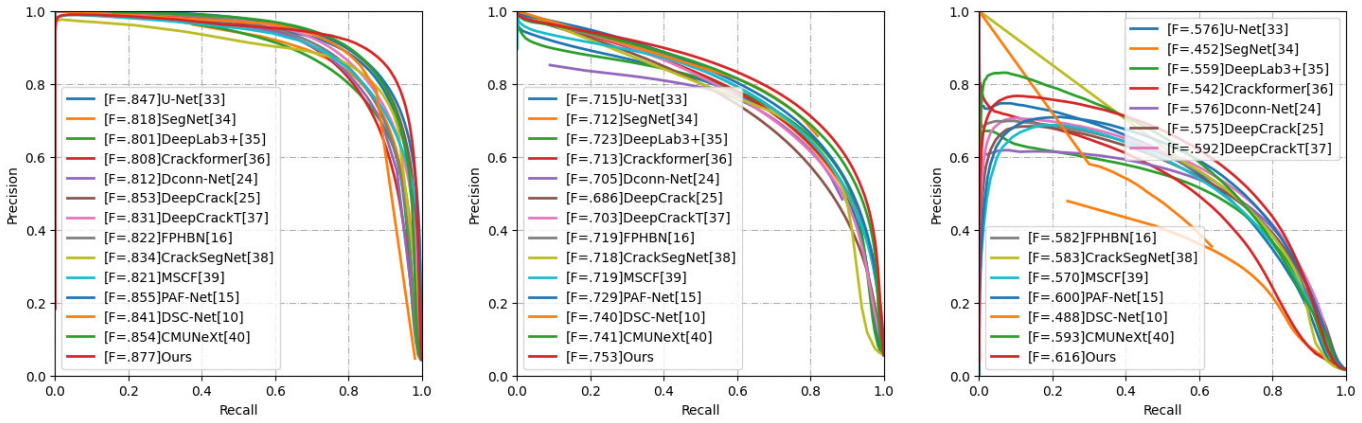

(a) (b) (c)

Fig. 11. Precision-Recall curves on the DeepCrack, Crack500 and CFD datasets respectively.

TABLE II
EVALUATION METRICS OF COMPARED METHODS ON CRACK500

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
U-Net [33]	0.682	0.752	0.715	0.747	0.966	0.866	0.761
SegNet [34]	0.692	0.734	0.712	0.742	0.967	0.873	0.767
DeepLab3+ [35]	0.681	0.769	0.723	0.741	0.967	0.871	0.766
Crackformer [36]	0.684	0.746	0.713	0.747	0.966	0.863	0.760
Dconn-Net [24]	0.671	0.742	0.705	0.732	0.966	0.857	0.754
DeepCrack [25]	0.661	0.713	0.686	0.710	0.964	0.840	0.742
DeepCrackT [37]	0.673	0.735	0.703	0.737	0.965	0.857	0.753
FPHBN [16]	0.698	0.742	0.719	0.746	0.967	0.862	0.764
CrackSegNet [38]	0.677	0.764	0.718	0.743	0.967	0.869	0.763
MSCF [39]	0.695	0.745	0.719	0.754	0.967	0.863	0.764
PAF-Net [15]	0.703	0.757	0.729	0.759	0.968	0.869	0.770
DSC-Net [10]	0.724	0.756	0.740	0.757	0.970	0.869	0.778
CMUNeXt [40]	0.705	0.780	0.741	0.770	0.970	0.874	0.778
Ours	**0.726**	**0.783**	**0.753**	**0.775**	**0.971**	**0.883**	**0.787**

TABLE III
EVALUATION METRICS OF COMPARED METHODS ON CFD

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
U-Net [33]	0.533	0.626	0.576	0.610	0.986	0.801	0.695
SegNet [34]	0.440	0.630	0.518	0.570	0.980	0.808	0.664
DeepLab3+ [35]	0.491	0.649	0.559	0.580	0.982	0.818	0.685
Crackformer [36]	0.509	0.580	0.542	0.561	0.983	0.785	0.677
Dconn-Net [24]	0.502	**0.677**	0.576	0.584	0.984	0.833	0.694
DeepCrack [25]	0.516	0.651	0.575	0.593	0.984	0.820	0.694
DeepCrackT [37]	0.530	0.670	0.592	0.606	0.985	0.830	0.703
FPHBN [16]	0.523	0.656	0.582	0.598	0.985	0.811	0.698
CrackSegNet [38]	0.538	0.636	0.583	0.595	0.985	0.814	0.698
MSCF [39]	0.515	0.639	0.570	0.593	0.985	0.807	0.692
PAF-Net [15]	0.549	0.661	0.600	0.620	0.986	0.816	0.707
DSC-Net [10]	0.466	0.512	0.488	0.506	0.983	0.751	0.653
CMUNeXt [40]	0.561	0.630	0.593	0.613	0.986	0.800	0.704
Ours	**0.576**	0.662	**0.616**	**0.632**	**0.986**	**0.827**	**0.715**

Table III show that the optimal ODS result of 0.616 is obtained when the threshold is set to 0.36. SegNet has the worst performance in this dataset. Dconn-Net obtains the best recall value, indicating that it utilizes the connectivity between pixels to extract crack information as much as possible.

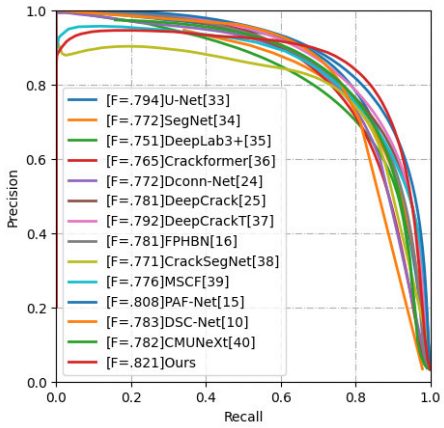

Fig. 12. Precision-Recall curves on the DCD dataset.

4) Results on DCD: As shown in Fig. 12, our method achieves the best performance on DCD mixed dataset, which indicates the better robustness and accuracy of our method in complex cracks and long cracks. The quantitative experiments in Table IV show that the optimal ODS result of 0.821 is obtained when the threshold is set to 0.24. The OIS value of CMUNeXt is slightly higher than our method, while the ODS value is lower than our method, indicating that CMUNeXt may lack some degree of generalization. The performance of our method is more stable.

5) Results on EdmCrack600 [32]: As shown in Fig. 13, our method achieves the best performance on EdmCrack600, but the overall performance is lower than that of other datasets. This dataset is more complex and involves various real-world environmental scenarios. The quantitative experiments in Table V show that the optimal ODS result of 0.541 is obtained when the threshold is set to 0.45. The ODS value of Dconn-Net is the closest to our method, indicating that Dconn-Net can be more sensitive to cracks when the background and cracks are significantly different, and U-Net with U-shape structure also performs well.

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TABLE IV
EVALUATION METRICS OF COMPARED METHODS ON DCD

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
U-Net [33]	0.824	0.765	0.794	0.809	0.987	0.880	0.822
SegNet [34]	0.767	0.777	0.772	0.802	0.985	0.885	0.807
DeepLab3+ [35]	0.741	0.762	0.751	0.788	0.983	0.876	0.792
Crackformer [36]	0.783	0.748	0.765	0.786	0.985	0.866	0.802
Dconn-Net [24]	0.780	0.764	0.772	0.791	0.985	0.873	0.807
DeepCrack [25]	0.775	0.788	0.781	0.805	0.985	0.890	0.813
DeepCrackT [37]	0.789	0.795	0.792	0.807	0.986	0.894	0.821
FPHBN [16]	0.787	0.774	0.781	0.793	0.986	0.884	0.813
CrackSegNet [38]	0.766	0.777	0.771	0.826	0.985	0.884	0.806
MSCF [39]	0.769	0.783	0.776	0.797	0.985	0.888	0.809
PAF-Net [15]	0.811	0.806	0.808	0.825	0.987	0.900	0.833
DSC-Net [10]	**0.836**	0.736	0.783	0.736	0.986	0.866	0.815
CMUNeXt [40]	0.790	0.775	0.782	**0.827**	0.986	0.884	0.814
Ours	0.820	**0.821**	**0.821**	0.825	**0.988**	**0.907**	**0.842**

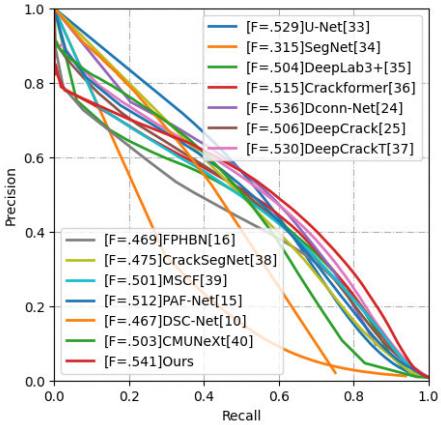

Fig. 13. Precision-Recall curves on the EdmCrack600 dataset.

TABLE V
EVALUATION METRICS OF COMPARED METHODS ON EDMCRACK600

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
U-Net [33]	**0.546**	0.514	0.530	0.544	0.992	0.755	0.676
SegNet [34]	0.337	0.297	0.315	0.303	0.989	0.646	0.588
DeepLab3+ [35]	0.458	0.560	0.504	0.438	0.991	0.777	0.664
Crackformer [36]	0.472	0.576	0.519	0.516	0.992	0.771	0.671
Dconn-Net [24]	0.527	0.544	0.536	0.542	0.992	0.770	0.679
DeepCrack [25]	0.474	0.544	0.506	0.522	0.991	0.770	0.665
DeepCrackT [37]	0.489	0.579	0.530	0.539	0.992	0.787	0.676
FPHBN [16]	0.408	0.551	0.469	0.384	0.990	0.772	0.648
CrackSegNet [38]	0.365	0.631	0.462	0.473	0.989	0.799	0.650
MSCF [39]	0.459	0.552	0.501	0.518	0.989	0.811	0.655
PAF-Net [15]	0.485	0.543	0.512	0.535	0.991	0.769	0.668
DSC-Net [10]	0.527	0.420	0.467	0.416	0.992	0.708	0.648
CMUNeXt [40]	0.513	0.493	0.503	0.525	0.992	0.737	0.664
Ours	0.507	**0.579**	**0.541**	**0.558**	**0.992**	**0.787**	**0.681**

6) Results on DCCE: Our objective is to investigate the model performance under various real-world pavement conditions, and therefore it is not necessary to test all comparative methods. As shown in Table VI, our method achieves a higher ODS value of 0.667 compared to other methods,

TABLE VI
EVALUATION METRICS OF COMPARED METHODS ON DCCE

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
Unet [33]	0.557	0.628	0.590	0.456	0.960	0.802	0.689
FPHBN [16]	0.576	0.662	0.616	**0.549**	0.962	0.819	0.703
Crackformer [36]	0.547	0.581	0.564	0.527	0.959	0.779	0.675
Dconn-Net [24]	0.622	0.650	0.636	0.524	0.967	0.806	0.716
Ours	**0.669**	**0.666**	**0.667**	0.512	**0.969**	**0.825**	**0.735**

indicating superior performance. The U-Net method shows slightly lower performance. The result of our method does not fall completely below those obtained on other independent datasets, suggesting that our method maintains a certain level of effectiveness across different pavement conditions. Therefore, it is reasonable to predict that in practical engineering scenarios, our method should not be entirely ineffective and should exhibit a significant degree of processing capability. As shown in Fig. 14, we present the visualization results for different pavement conditions. We find that our method still has some shortcomings, particularly in shadow and high exposure situations. This is an area we will continue to research.

7) Areas of Improvement: Our method performs well compared to other methods on six datasets, but it is undeniable that our approach still suffers from some issues. As shown in the second and third rows of Fig. 9, our method exhibits minor flaws when it extracts complex cracks. We consider that our network tends to focus relatively more on long cracks, and using only ESP may not fully support the detection of other types of cracks. Additionally, we utilize pixel connectivity in our loss module. When this network deals with very small and isolated cracks, the relationship of pixel connectivity may be influenced by background pixels, leading to detection failures. To illustrate potential issues more comprehensively, we visualize the detection results that are less than satisfactory under various real-world conditions in the EdmCrack600 dataset, as shown in Fig. 15. Shadows from trees produce two shapes: one with a mesh-like pattern and one without, in the first row of Fig. 15. Some pixels of the mesh-like shadow are similar to the pavement, which our model may classify as non-cracks due to connectivity learning. Pixels of the non-mesh-like shadow differ significantly from the pavement, potentially leading the connectivity loss module to misclassify them as cracks during learning. Similarly, in the second row of Fig. 15, due to the distinct variations in pixels at the edges of puddles, snow accumulation, and pavement, a edge line may appear in the image, potentially causing some extent of false detection. Since we employ connectivity modeling, which emphasizes relationships between pixels, there could be cases where crack pixels closely resemble pavement pixels under high-exposure conditions, potentially leading to missed detections, as seen the third and fourth rows of Fig. 15.

8) Running Efficiency: As shown in Table VII, compared to other methods, our method is thrid only to Dconn-Net and CMUNeXt in terms of $FLOPs$, but our accuracy is higher and the number of parameters is relatively lower. The lowest number of parameters is CMUNeXt, but our number of parameters

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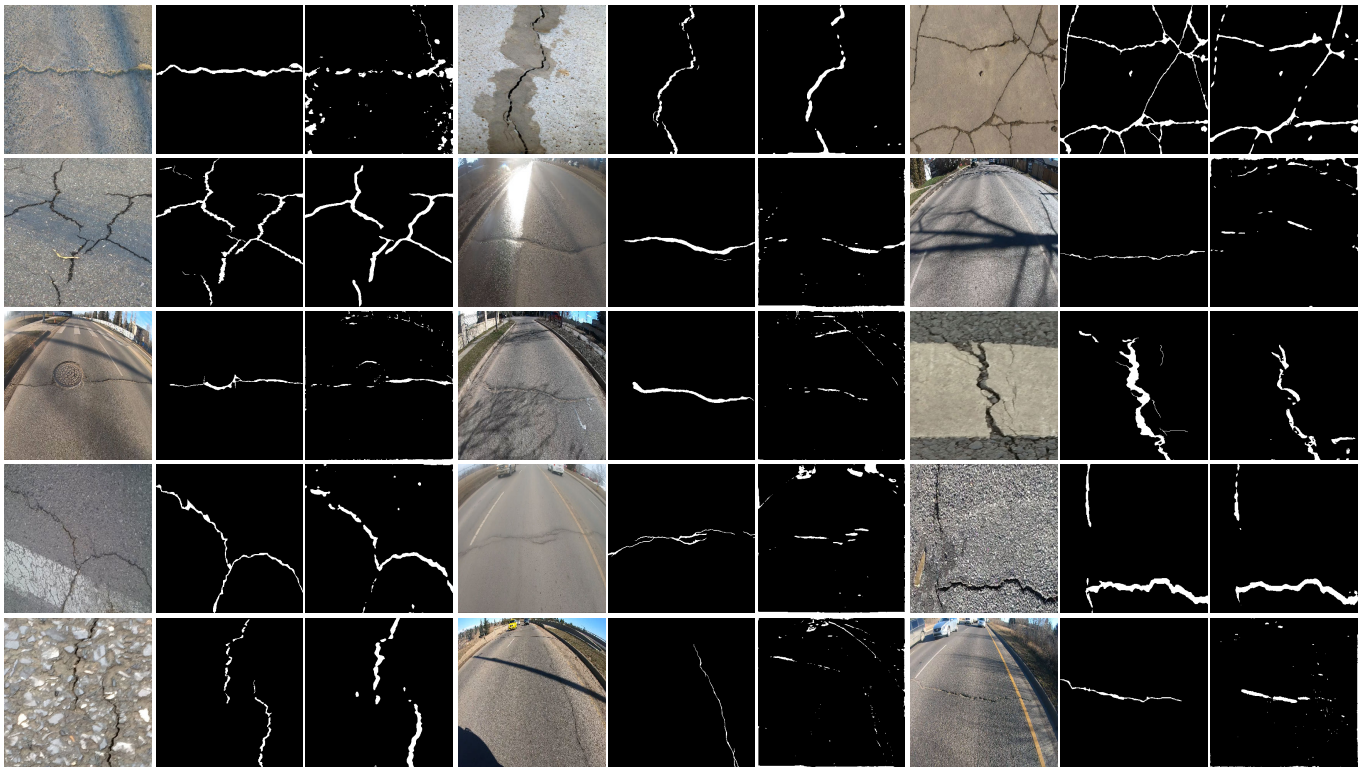
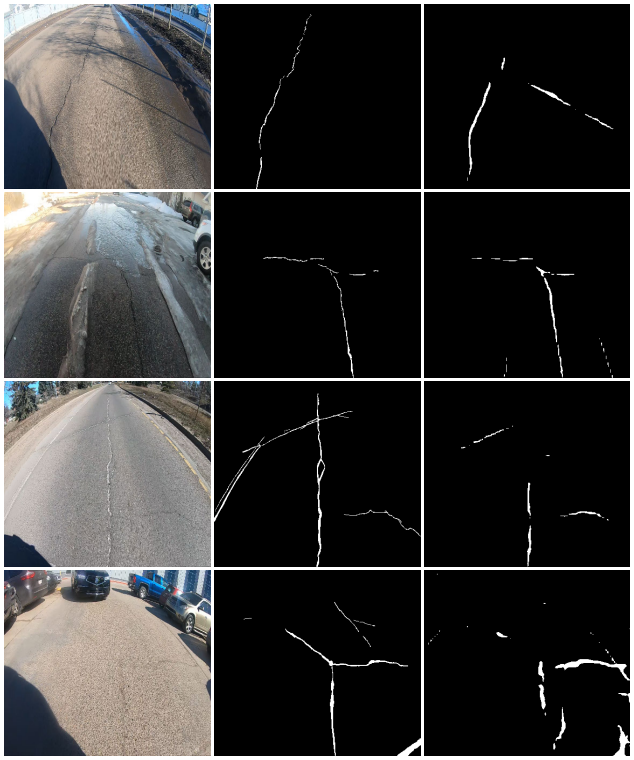


(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 14. The visualization of detection results of proposed methods on DCCE dataset. (a)(d)(g) Raw Image; (b)(e)(h) Ground truth; (c)(f)(i) Ours.



(a) (b) (c)

Fig. 15. The visualization of detection results of proposed methods on EdmCrack600. (a) Raw Image; (b) Ground truth; (c) Ours.

is thrid only to CMUNeXt and Crackformer. In terms of FPS, our method is not overly prominent, but considering several other aspects, our model obtains a relatively balanced model performance improvement.

TABLE VII

THE $Flops$, $Params$ AND FPS OF ALL METHODS ON TEST DATASET OF DEEPCRACK

Methods	$FLOPs$	$Params$	FPS
U-Net [33]	54.68G	31.03M	14
SegNet [34]	33.46G	24.94 M	18
DeepLab3+ [35]	40.10G	29.44M	11
Crackformer [36]	20.49G	4.55M	3
Dconn-Net [24]	6.21G	25.49M	16
DeepCrack [25]	20.09G	14.72M	25
DeepCrackT [37]	40.12G	29.45M	16
FPHBN [16]	22.51G	14.98M	24
CrackSegNet [38]	57.55G	14.84M	14
MSCF [39]	35.67G	27.52M	12
PAF-Net [15]	45.91G	12.62M	13
DSC-Net [10]	45.91G	12.61M	14
CMUNeXt [40]	7.38G	3.14M	10
Ours	**11.85G**	**6.15M**	**12**

F. Ablation Experiments

1) Ablation Experiments of the MDECM: The MDECM consists of the directional mixed strip convolution (DMSC) and the effective dilated spatial pyramid (ESP). To prove the effectiveness of the overall module, we substitute these two parts separately into the MDECM of the backbone network without the multi-scale fusion and the directional connectivity loss, which more intuitively illustrates the capabilities of these two modules. The Table VIII shows that DMSC and ESP alone have some extraction capability and the extraction capabilities are relatively close. Thus, when fusing the two parts, it does not occur that one pulls the other down. Meanwhile, the

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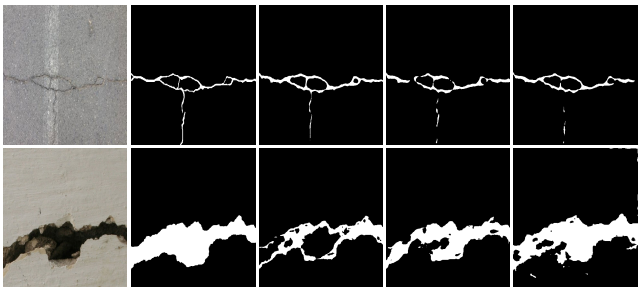
TABLE VIII
ABLATION EXPERIMENTS OF THE MDECM ON DEEPCRACK

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
DeepCrack [25]	0.855	0.850	0.853	0.878	0.987	0.922	0.865
DMSC	0.855	0.855	0.855	0.879	0.987	0.924	0.867
ESP	0.849	0.856	0.852	0.877	0.987	0.924	0.864
MDECM	**0.864**	**0.868**	**0.866**	**0.886**	**0.988**	**0.931**	**0.876**



(a) (b) (c) (d) (e)

Fig. 16. The visualization results of the Ablation results of the DMSC, ESP and DeepCrack. (a) Raw Image; (b) Ground truth; (c) DMSC; (d) ESP; (e) DeepCrack.

TABLE IX
ABLATION EXPERIMENTS OF THE MFA ON DEEPCRACK

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
(MFA-A, MFA-A)	0.851	0.865	0.858	0.881	0.987	0.929	0.869
(MFA-A, MFA-B)	**0.870**	0.856	0.863	0.888	0.988	0.925	0.873
(MFA-B, MFA-B)	0.866	0.862	0.864	0.886	0.988	0.928	0.874
(MFA-B, MFA-A)	0.866	**0.872**	**0.869**	**0.891**	**0.988**	**0.933**	**0.878**
ECA [29]	0.864	0.859	0.861	0.885	0.988	0.926	0.872
CoordAtt [31]	0.865	0.854	0.859	0.885	0.988	0.924	0.870
ECA+CoordAtt	0.863	0.863	0.863	0.886	0.988	0.924	0.873
MFA-Single	0.870	0.863	0.867	0.887	0.989	0.921	0.876
MFA-B-3	0.867	0.848	0.858	0.885	0.988	0.921	0.869
MFA-B-2+3	0.867	0.865	0.866	0.889	0.988	0.929	0.876
MFA-B-1+2+3	0.865	0.851	0.858	0.887	0.988	0.922	0.869

TABLE X
ABLATION EXPERIMENTS OF THE CONNECTIVITY LOSS MODULE ON DEEPCRACK

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
L_{c_side}	**0.870**	0.867	0.868	0.897	0.988	0.930	0.878
L_{c_side} + L_{conn_side}	0.852	0.859	0.856	0.888	0.987	0.926	0.867
$\boldsymbol{L_{conn_side}}$	0.863	**0.891**	**0.877**	**0.899**	**0.989**	**0.942**	**0.884**

experimental results also show that combining the two parts can improve the performance. Additionally, to verify that DMSC is more sensitive to long strip cracks, we compare DMSC with DeepCrack which uses square convolution to construct the backbone network. In Table VIII, the ODS values of the two methods are similar, indicating that both strip and square convolutions have some effect, but they may differ in the aspects of interest. In Fig. 16, the results in the first row show that DMSC is better at detecting strip cracks, and the results in the second row indicate that the DMSC also weakly detects other cracks, thus also highlighting the necessity of ESP.

2) Ablation Experiments of MFA: The MFA includes two methods, MFA-A and MFA-B. MFA-A focuses on the global position and direction relations, and MFA-B further focuses on the mutual relations between channels. Therefore, we use the two methods at different positions of feature fusion ($f_{(S2,S3)}$, $f_{(S3,S4,S5)}$). As shown in Table IX, it is easy to see that MFA-B has great performance regardless of the position. MFA-A is more friendly to high-level semantic information, which does not require additional operations to explore channel relations. There is still a large amount of detail information in the shallow layers, which needs to be filtered through the relations between channels. Therefore, this structure of (MFA-B, MFA-A) achieves the best performance. To explain the rationality of using the MFA module, we gradually incorporate the MFA-B module into the side outputs of the our original network structure in the first three stages. In Table IX, we observe that varying the number of MFA modules yields different results, and the impact is not as positive as in the original approach. Therefore, we fully utilize the MFA modules that we design. Meanwhile, to validate the effectiveness of the proposed MFA module, we replace the MFA module with ECA [29], CoordAtt [31] and parallel ECA+CoordAtt in the network for comparison experiments. In Table IX, the parallel ECA+CoordAtt performs better than using one of the modules alone, indicating that the parallel mode can improve the performance of the network. Corresponding to our proposed module, the MFA-Single module (an MFA module without ECA) performs better than CoordAtt and ECA. The effectiveness of this module is further enhanced by adding ECA to form parallel branches. Therefore, this not only demonstrates the effectiveness of parallel branch in our proposed MFA module but also illustrates the superior feature fusion capability of MFA branch.

3) Ablation Experiments of the Connectivity Loss: Our method introduces directional connectivity loss to strengthen the learning ability of the network, which needs to consider the way the connectivity loss is used. Therefore, in the side loss, it is selected whether to use the directional connectivity loss or the standard binary cross-entropy loss for the experiment. The Table X shows that using directional connectivity loss on the side is the best method, and Using the standard binary loss function (L_{c_side}) only on the side is also inadequate because the connectivity of each side is not effectively utilized. The effect is the worst when using two losses on the side (L_{c_side} + L_{conn_side}), because there is a competitive relationship between the features provided by the backbone network for the two losses, and the obtained feature results do not meet the requirements of the loss function. Meanwhile, the directional connectivity loss itself has a redundant relationship between the eight-channel connectivity loss and the bilateral voting loss, and the coefficients α and β are set to explore a reasonable loss method. The Table XI indicates that the performance of the model is optimal with $\alpha = 0.8$ and $\beta = 0.2$.

4) Ablation Experiments of the Modules: We use the DeepCrack backbone network as the baseline. In Table XII, the performance of MDECM and Conn_Loss is similar. However,

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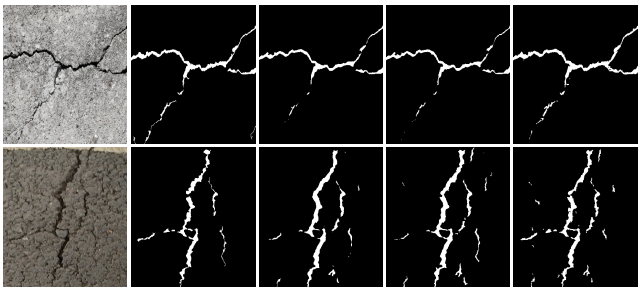
TABLE XI
ABLATION EXPERIMENTS OF THE L_{conn_map}/L_{bv_map} MODULE ON DEEPCRACK

α/β	P	R	ODS	OIS	PA	MPA	$MIoU$
0.9/0.1	0.862	0.874	0.868	0.891	0.988	0.934	0.877
0.8/0.2	0.863	**0.891**	**0.877**	**0.899**	**0.989**	**0.942**	**0.884**
0.7/0.3	**0.864**	0.874	0.869	0.890	0.988	0.934	0.878
0.6/0.4	0.849	0.839	0.844	0.881	0.986	0.916	0.858
0.5/0.5	0.757	0.785	0.771	0.864	0.980	0.877	0.803

TABLE XII
ABLATION EXPERIMENTS OF THE PROPOSED MODULE ON DEEPCRACK

Methods	P	R	ODS	OIS	PA	MPA	$MIoU$
MDECM	0.864	0.868	0.866	0.886	0.988	0.931	0.876
MFA	**0.871**	0.867	0.869	0.892	0.989	0.930	0.878
Conn_Loss	0.853	0.878	0.866	0.895	0.988	0.936	0.875
MDECM+MFA	0.866	0.872	0.869	0.891	0.988	0.933	0.878
MDECM+Conn_Loss	0.836	0.847	0.842	0.871	0.986	0.920	0.856
MFA+Conn_Loss	0.852	0.869	0.860	0.890	0.988	0.931	0.871
MDECM+MFA+Conn_Loss	0.863	**0.891**	**0.877**	**0.899**	**0.989**	**0.942**	**0.884**



(a) (b) (c) (d) (e)

Fig. 17. The visualization results of the Ablation results of the proposed methods. (a) Raw Image; (b) Ground truth; (c) MDECM; (d) MDECM+MFA; (e) MDECM+MFA+Conn_Loss.

the performance of MEDCM+Conn_Loss is lower than either of them. We consider that when the backbone network does not integrate and utilize the feature information reasonably, subsequent Conn_Loss cannot effectively fcous on the information of neighboring pixels. The highest ODS of MFA module alone is that it may have used the information extracted by square convolution in the backbone network to the maximum extent, and this does not mean that their overall performance is the best, because the feature information extracted by square convolution is limited, resulting that subsequent operations can not get more information from it. For example, MFA+Conn_Loss leads to a decrease in ODS value. While the ODS and $MIOU$ values of MFA+MDECM remain the same compared to MFA, the R value increases slightly, which seems to indicate a similar performance between the two. However, MFA+MDECM stores more information about crack features. By adding Conn_Loss to this, the accuracy of the model is greatly improved. Thus, each module possesses distinct characteristics, and leveraging them effectively maximizes their potential advantages. The ODS value of MDECM alone is 0.866, which increases to 0.869 with the addition of MFA. Further incorporating Conn_Loss raises the ODS value to 0.877. Fig. 17 also shows this process, and crack detection results are gradually getting better with the addition of modules.

V. CONCLUSION

We design the multi-directional strip convolutional module as the basic module of the backbone network to capture crack features of complex topologies and provide better features for subsequent methods. Besides, we build multi-scale fusion attention to generate the long range of directional position information which guides to fuse feature channels, and introduce the directional connectivity loss to exploit the relationship between neighboring pixels which refines the details of cracks and improves detection performance. Therefore, a directional connectivity feature enhancement network is proposed, which contributes greatly to improvements in performance.

In the future, we will further investigate connectivity methods that enhance the performance of the backbone network to better suit connectivity, and intend to fully leverage connectivity information and consider whether the distance between neighboring pixels affects crack feature extraction. Furthermore, we will thoroughly consider the impact of various real-world environments and further study pavement cracks under high exposure and shadow conditions, aiming to construct models that are suitable for practical engineering problems.

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